



BRIGHTLEARN

BRIGHT COFFEE SHOP SALES ANALYSIS

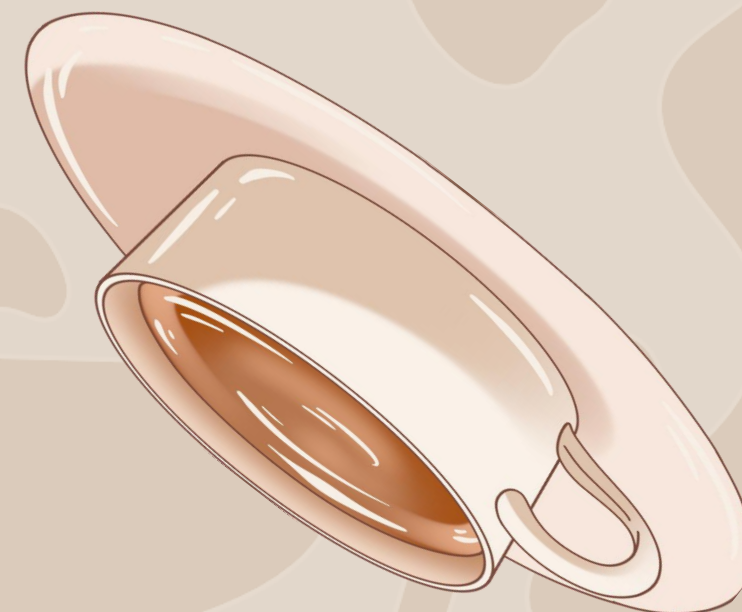
Turning Sales Data into Actionable Business Insights
Presented by: Ntshovelo Hlongwane | Programme: Data
Analytics | BrightLearn | 2026
Tools: Databricks | SQL | Excel | Power BI | Looker
Studio | Lovable





PROJECT INTRODUCTION

This project focuses on analysing transactional sales data for Bright Coffee Shop. Data cleaning and transformation is performed using SQL in Databricks, followed by exploratory analysis, pivot tables, and data visualisation. Goal: Transform raw sales data into actionable business insights that support informed decision-making.





PROJECT OBJECTIVES

- Analyse transactional sales data for Bright Coffee Shop
 - Identify highest revenue and best-selling products
- Determine peak sales periods and patterns
 - Compare weekday vs weekend sales
- Create KPIs to measure and track sales performance
- Develop data-driven recommendations for the business





BUSINESS QUESTIONS

Analysis Guided by These Questions

1. Which products generate the most revenue?
2. Which products sell the most units?
3. What are the peak sales periods?
4. How does sales vary across days?
5. How does sales change over time?
6. What actionable steps can Bright Coffee Shop take?





DATA SOURCE

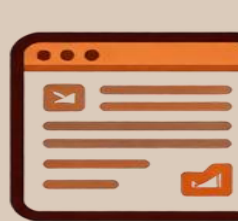
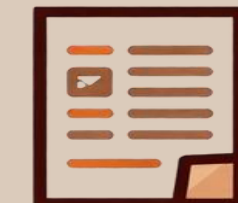
Bright Coffee Shop transactional sales data. Fields: transaction date/time, quantity, store, product, product type/detail, and unit price. Data converted from Excel to CSV and loaded into Databricks. Pipeline: Raw Dataset → Cleaning → Transformation → Analysis → Visualisation.





TOOLS & TECHNOLOGIES

Databricks: Storage, processing & SQL-based analysis. SQL: Data cleaning, transformation & querying. Excel: Pivot tables & exploratory charts. Power BI, Looker Studio & Lovable: Interactive dashboards and reports. GitHub: Project documentation. Together, these tools support the full end-to-end analytics workflow.





DATA CLEANING & TRANS- FORMATION



Key data preparation steps applied to the raw dataset:

- Standardised unit_price field and converted comma decimals to numeric values
- Calculated total_amount = unit_price × quantity
- Extracted Day Name, Month Name, Year, and Hour of the Day
- Classified each transaction as Weekday or Weekend
- Created 3-hour Time_Bucket intervals and broader Time_Period categories
- Ensured clean, reliable data ready for analysis





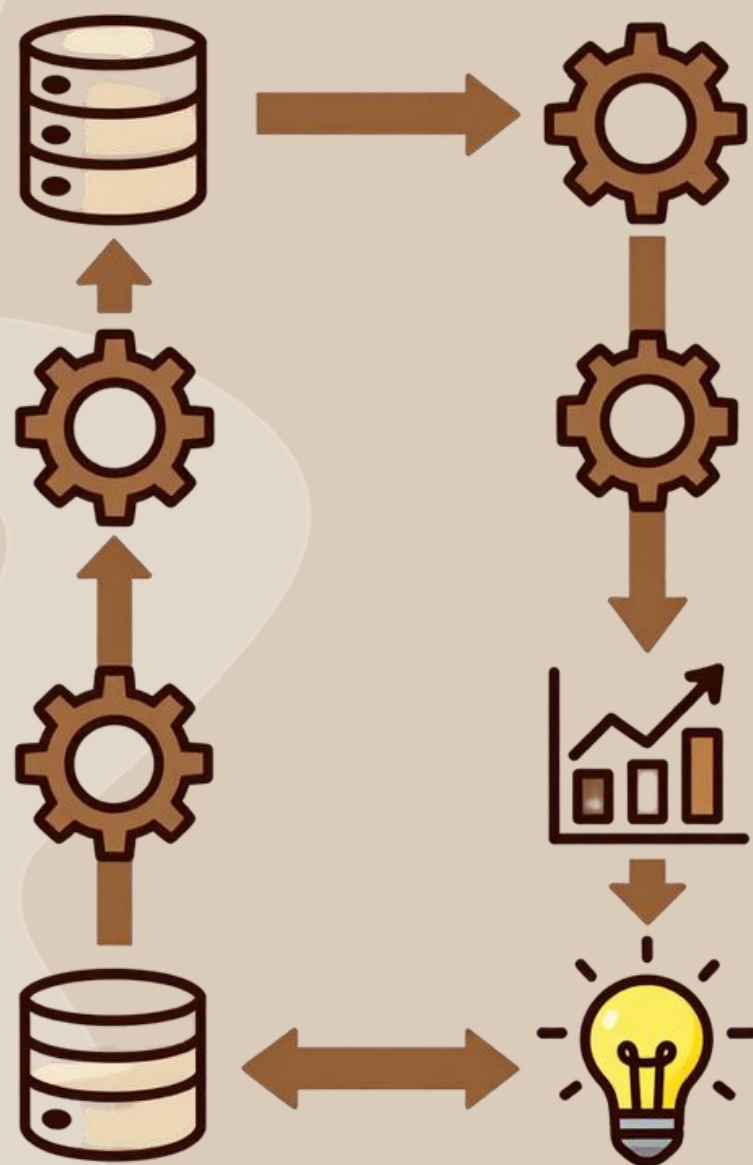
CLEANED DATASET

The analytical dataset retains original fields plus calculated ones: transaction_id, transaction_date, time_of_day, transaction_qty, store_id, product_id, product_type, product_detail, unit_price_standardized, total_amount, Day_Name, Month_Name, Transaction_Year, Transaction_Day, Hour_of_the_Day, Day_Classification, Time_Bucket, and Time_Period — enabling multi-dimensional sales analysis.





ANALYTICAL WORKFLOW



The project followed a structured end-to-end analytics process:
RAW DATA → DATA CLEANING →
DATABRICKS + SQL → CALCULATED FIELDS
→ EXPLORATORY ANALYSIS → PIVOT
TABLES → DASHBOARDS → INSIGHTS →
RECOMMENDATIONS
Each stage builds on the previous,
transforming raw transactional sales
data into clear, actionable business
insights for Bright Coffee Shop.



KEY PERFORMANCE INDICATORS



- **Total Revenue:** $\text{SUM}(\text{total_amount})$ — overall sales value generated.
- **Total Units Sold:** $\text{SUM}(\text{transaction_qty})$ — volume of items purchased.
- **Total Transactions:** $\text{COUNT}(\text{transaction_id})$ — number of sales events.
- **Average Transaction Value:** $\text{Total Revenue} \div \text{Total Transactions}$. **Top-Selling Product** by quantity sold. **Top Revenue-Generating Product** by total amount. **Peak Sales Period** by highest revenue interval.



PRODUCT PERFORMANCE ANALYSIS



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Revenue and quantity sold were analysed by product type and individual product. The top 10 products were ranked by both total revenue and units revenue contribution. Bar charts highlight the highest-performing products and product types, revealing which items drive the most sold, enabling a direct comparison of sales volume versus value for Bright Coffee Shop.





PRODUCT PERFORMANCE FINDINGS

Highest Revenue Product: Sustainably Organic
Lg , R21 151.75

Best-Selling Product: Earl Grey Rg, 4708
units

Highest-Performing Product Type: Barista
Espresso, R 91 406.20

Observation: Significant revenue contributors
identified, with clear opportunities to promote
top-performing products and address
underperforming categories.





TIME-BASED SALES ANALYSIS

Revenue and quantity sold analysed by Time_Bucket (3-hour intervals) to identify peak and off-peak trading windows. Hour-of-day breakdowns reveal granular demand patterns, while Time_Period categories (Morning, Midday, Afternoon, Evening) highlight broader daily rhythms. Line and bar charts visualise sales peaks and troughs throughout the day.





TIME-BASED FINDINGS

Highest-Performing Period: 09:00 – 11:59 — Revenue
R220 12.0 | Units Sold: 66 118

Lowest-Performing Period: 18:00 – 20:59 — Revenue
R65 68.58 Units Sold: 20 301

Observation: Peak demand windows present key opportunities for targeted promotions and optimised staffing. Slow periods should be leveraged with strategic campaigns to drive incremental revenue growth.



DAY & CALENDAR FINDINGS



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Highest-performing day: Monday — strongest revenue and unit sales recorded.

Lowest-performing day: Saturday — potential target for promotions.

Highest-performing month: June — peak seasonal demand identified.

Weekday revenue: R503 587.54 vs Weekend revenue: R195 224.79.

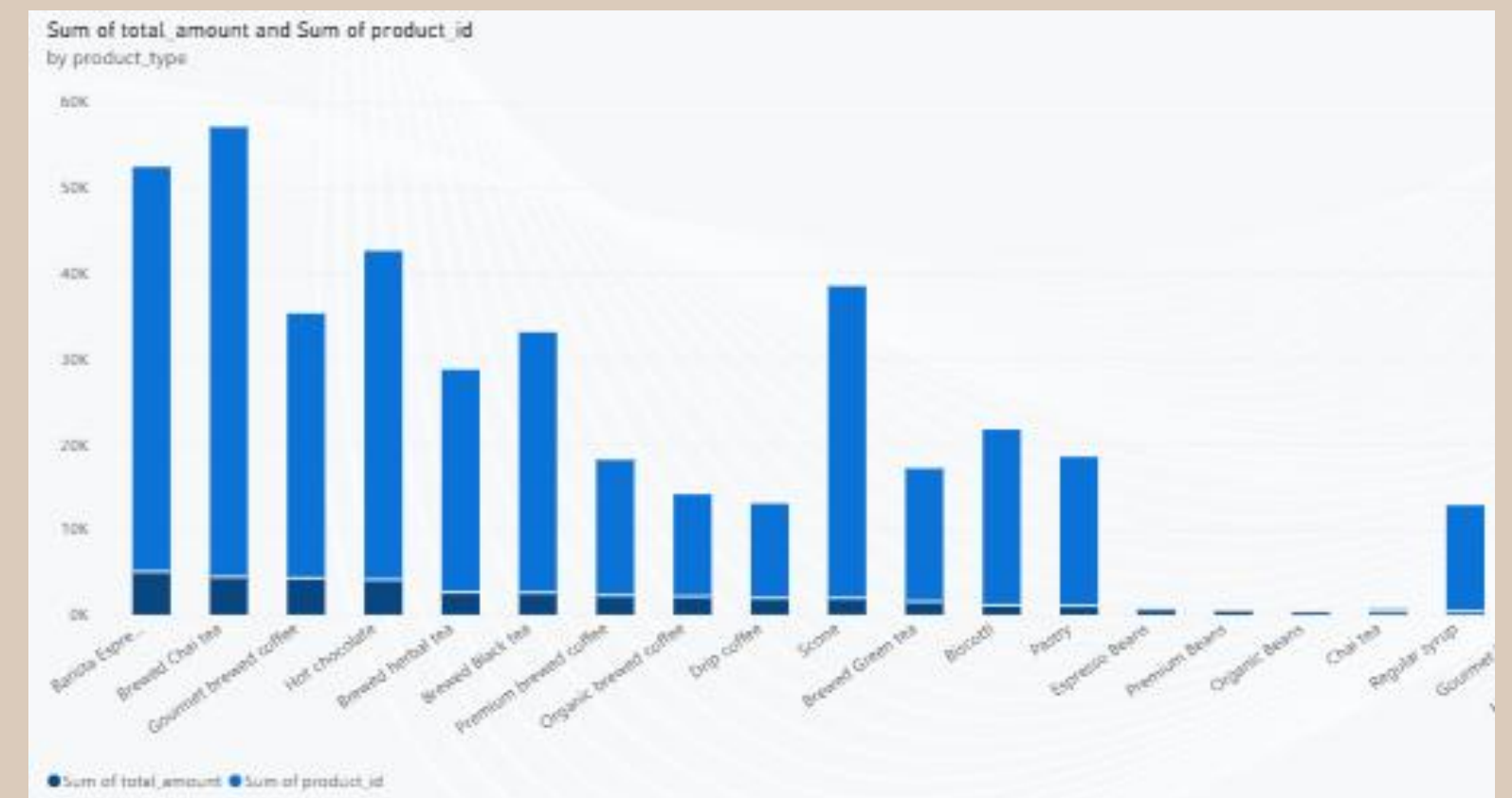
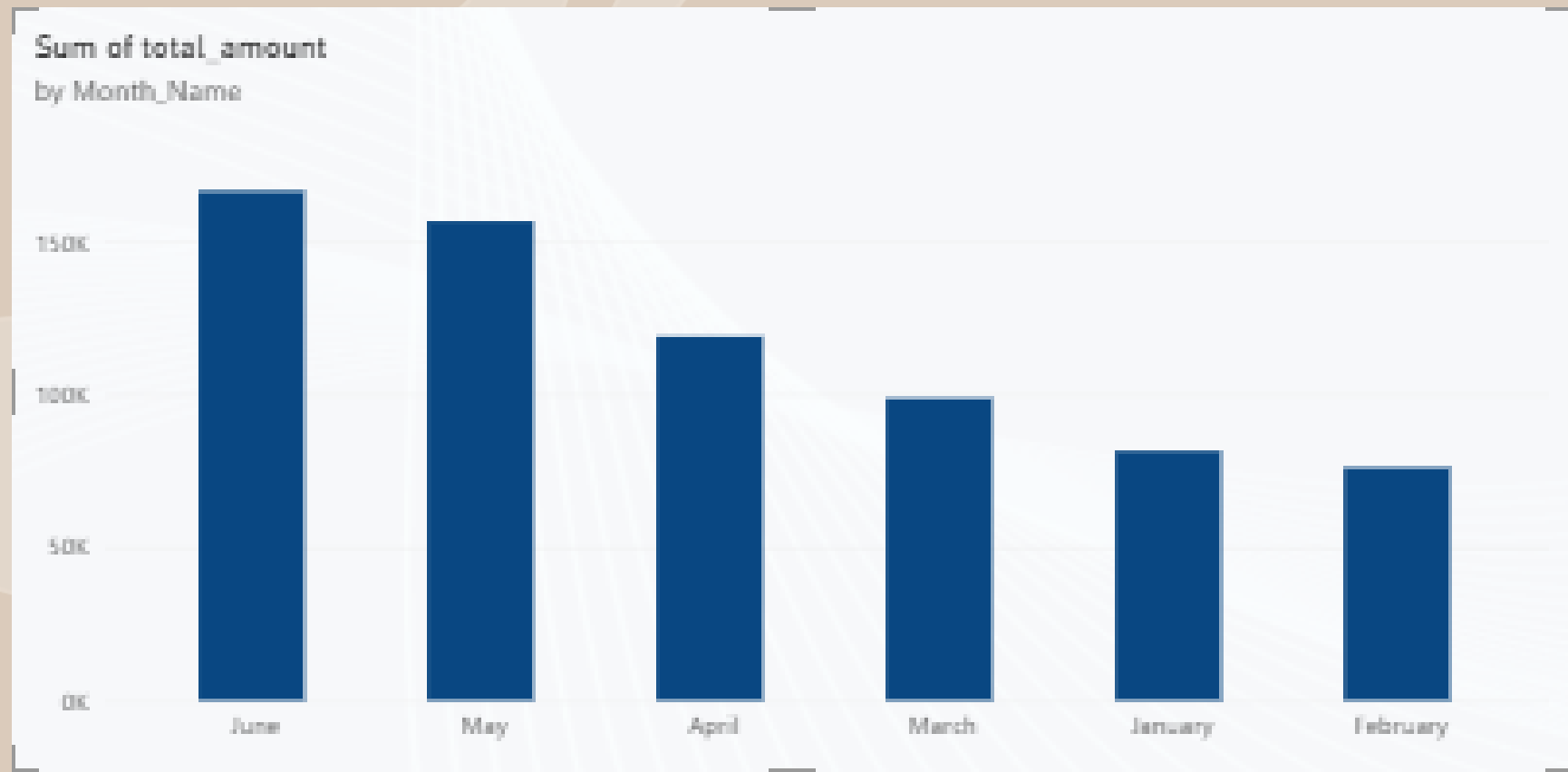
Observation: Demand is concentrated on specific days, highlighting clear marketing focus areas and opportunities to drive sales during slower periods.





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DASHBOARD SCREENSHOTS





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KEY INSIGHTS

- 01 — Total Revenue: R698 812.33
- 02 — Top Revenue Product: Sustainably Grown Organic LG, R21 151.75
- 03 — Highest Sales Volume Product: Earl Grey Rg, 4708 units
- 04 — Peak Sales Period: 09:00 – 11:59, R220 162.0
- 05 — Strongest Sales Day/Month: June





BUSINESS IMPLICATIONS

Stock Planning: Ensure top-selling products are well-stocked during identified peak periods to avoid lost sales. Marketing: Target promotions and campaigns during slow sales periods to drive traffic and boost revenue. Product Strategy: Prioritise high-revenue products in marketing campaigns and ensure prominent placement. Sales Planning: Use historical day and time patterns to guide staffing levels and stock management decisions.





RECOMMENDATIONS

1. Optimise stock for top products during peak sales times
2. Promote products during slow periods with targeted campaigns
3. Improve underperforming products via bundling and placement
4. Align marketing calendar with identified sales patterns
5. Monitor product sales regularly to detect shifts
6. Automate sales reporting for timely, ongoing insights





LIMITATIONS

This analysis is based on historical transactional sales data only. No customer demographic or retention data was available, limiting behavioural insights. Profitability analysis is constrained without cost data. Identified patterns reflect correlation, not causation. Recommendations should be considered alongside broader business context and strategy.





NEXT STEPS

1. Automate daily/weekly sales reporting for timely insights.
2. Compare performance across multiple store locations.
3. Incorporate customer-level data for behaviour analysis.
4. Develop loyalty programmes using purchase history data.
5. Use predictive analytics for demand forecasting.
6. Maintain continuously refreshed interactive dashboards.





THANK YOU.

